

Machine Learning-Driven Long/Short Pair Trading Strategies in Cryptocurrencies

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Abstract

Traditional trading strategies such as momentum-based or buy-and-hold, in most cases, failed to cater for more complex assets' interdependencies or non-linear trends witnessed across financial time series. The current study was aimed at designing and assessing ML and DL- driven long/short pairs trading strategies for cryptocurrencies, to enhance predictive precision and trading performance. The project data was extracted from Yahoo finance, leveraging on nine-years (9) of daily historical data for cryptocurrency assets (Bitcoin, Ethereum, Ripple, Litecoin, Cardano, Polkadot, Solana, Avalanche and Binance Coin) from 30th September 2016 to 29th September 2025. The project first filtered potential pairs with cointegration, then forecasted with LSTM, then used ensemble models for risk-adjusted allocation. On evaluation, baseline strategies (buy-and-hold, momentum and equal-weight) were used, alongside Sharpe ratio, drawdown, and cumulative return metrics. Incorporation of statistical co-integration tests, LSTM-based DL-models and ensemble ML techniques, the project was able to construct and further evaluate strategies with the ability to maximize returns as they minimize risk, and backtest the various strategies systematically.

Key words: Machine Learning (ML), Cryptocurrency Trading, Pairs Trading, LSTM Networks, Backtesting, Long/Short Strategy, Cointegration, Ensemble Models, Risk-Adjusted Returns, Deep Learning (DL)

1.0 Introduction

The rise in cryptocurrency markets has come with a mix of challenges and opportunities for quantitative traders. As opposed to foreign exchange markets or traditional equities, crypto markets are characterized with non-stationarity, extreme volatility and structural breaks, which often undermine efficient market behaviour and linear model assumptions (Qureshi; Eugene, Fama 383). Moreover, static approaches such as moving-average crossovers or buy-and-hold often fail to realize risk-adjusted consistent returns. This setback has motivated the essence of deep learning (ML) and machine learning (ML) techniques, with the ability to handle dynamic dependencies and non-linearity, common with digital assets.

Machine-learning-driven long/short pair trading is a market-neutral approach, as it builds on temporary mispricing witnessed across historically related assets (Stanciu). Cointegration, a theoretical foundation is this approach, aids in examining the existence of stationary linear combination between non-stationary price series.

Two asset prices P_{1t} and P_{2t} are cointegrated if a scalar β exists in which the spread defined as;

$$S_t = P_{1t} - \beta P_{2t} \quad (1)$$

And from (1), $S_t \sim I(0)$, even though the individual prices follow $\{P_{1t}, P_{2t}\} \sim I(1)$ processes. This aspect of stationarity implores mean-reverting behaviour of the series, which allows trader short on overvalued and long on undervalued assets, whenever deviations from equilibrium are witnessed (Piallini).

Nonetheless, the traditional cointegration tests (Johansen, Engle-Granger) works by assuming static relationships, which often become weak under sentimental-driven dynamics and regime shifts as in the case of crypts (Bilgili). Addressing this limitation, reveals the essence of Long Short-Term Memory (LSTM) networks, in a bid to include time-varying dependencies and forecasts, viable for future spreads.

Provided a sequence of spreads defined as $(S_{t-k}, \dots, S_t)$, a LSTM model learns a function

$$\widehat{S_{t-k}} = f_{LSTM}(S_{t-k:t}) \quad (2)$$

which captures both long and short-range temporal patterns, allowing for dynamic forecasting of divergence and convergence phases.

Moreover, the trading allocation problem is often formulated through ensemble learning. This approach includes a series of base learners such as Bagging, Linear Regression, Gradient Boosting

and Random Forests, for optimization of position sizing (Ferrouhi and Bouabdallaoui). The ensemble results in an allocation function defined as;

$$w = gensemble(\widehat{S}_{t+1}, \sigma_t, features_t) \quad (3)$$

which maximizes Sharpe ratio and minimizes portfolio volatility, as proposed in the current study.

1.2 Goals and Objectives

1.2.1 Goal:

The goal of this study was to design and assess ML and DL- driven long/short pairs trading strategies for cryptocurrencies, to enhance predictive precision and trading performance.

1.2.2 Objectives:

1. To undertake a detailed literature search on cointegration, statistical arbitrage and adoption of ML and DL approaches in pairs trading.
2. To develop ML (ensemble methods) and DL models (LSTM networks) to examine and exploit pair trading opportunities.
3. To backtest the realized strategies using historical data and evaluate performance of the different portfolios using Sharpe ratio, maximum drawdown, and profitability metrics.
4. To compare the effectiveness of DL and ML-based approaches with trading pairs for assessing improvements in risk management and predictive accuracies for investor insights.

1.3 Problem Statement:

Traditional trading strategies such as momentum-based or buy-and-hold, in most cases, failed to cater for more complex assets' interdependencies or non-linear trends witnessed across financial time series data (Fior). Moreover, these approaches also relied on linear models and often assumed constant asset return relationships. Nonetheless, financial markets have proved non-linear and complex, hence the need for more sophisticated approaches to take care of these dynamics and trends. The gap has further motivated the essence of both deep learning (DL) and machine learning (ML) methodologies, to aid in forecasting asset prices, to detect any anomalies and further aid in algorithmic trading strategies around more mature financial markets. Despite the rise in interest, the adoption of ML-driven long/short pair trading strategies in cryptocurrencies has been unexplored largely (Alao). Statistical arbitrage approaches which exploit cointegration between various asset pairs, have been examined across global markets, but less studies in crypto markets. Furthermore, the

advancements in deep learning which include Long Short-Term Memory (LSTM) networks alongside other ML approaches like ensemble methods, provide viable room to better portfolio trading performance and their predictive accuracy (Shah, Vaidya, and Shah 200111). Nonetheless, there is limited empirical research on how such methods can be effectively applied to cryptocurrency portfolios, especially around markets experiencing variation in structural characteristics and liquidity profiles.

2.0 Literature Review

The evolution of deep learning (DL) and machine learning (ML) approaches have enhanced the existing traditional strategies to algorithmic trading and financial forecasting. Cryptocurrencies, as part of the emerging markets are still under research, owing to their non-stationarity, high volatility and limited data available (Ze, Yang, and Ooi 255). Their non-linear and dynamic nature of such markets makes the traditional methods of buy-and-hold, momentum and equal-weight less viable (Fior). In this case, there is need for more data-driven approaches, hence the essence for ensemble learning and LSTM networks to better accuracy and risk management across long/short trading pair frameworks (Shah, Vaidya, and Shah 200111).

The review examines three key areas: traditional trading strategies and their limitations, adoption of statistical arbitrage and cointegration across emerging markets and DL and ML as new proposals for the enhancement of profitable trading approaches for cryptocurrencies.

2.1 Traditional Trading strategies

Conventional trading approaches such as moving-average crossovers, buy-and-hold and momentum are coined in the assumption of linear dependencies and market efficiency (Acharya). Nonetheless, these emerging markets are also characterized by structural breaks, time-varying volatility and non-stationarity, which are notable for reducing robustness of these traditional approaches (Mikolajek-Gocejna).

According to Lo and MacKinlay, most financial returns often deviate from normality, informing invalidated linear assumptions intrinsic around most econometric models therein. Moreover, Fama's proposed Efficient Market Hypothesis (EMH), despite being a viable foundation, has faced a series of empirical setbacks on behavioural inconsistencies and market anomalies (Fama 383). On the flipside, DL and ML models have proved their capabilities in detection of latent patterns, capturing non-linearities and adjusting to regime changes across volatile and high-frequency data (Hu et.al). Furthermore, the limitation of traditional frameworks also informs the essence of advanced data-

driven methodologies, with the ability to learn complex dynamics, especially around cryptocurrency markets, associated with microstructural noise and heteroskedasticity.

2.2 Cointegration and Statistical Arbitrage across financial markets

Statistical arbitrage, is mostly embedded on cointegration analysis, suitable for examining mean-reverting relationships between different asset pairs (Gatev, Goetzmann, and Rouwenhorst 797). The Engle–Granger (1987) approach, a two-step method and Johansen’s (1988) multivariate approaches are key tools employed for testing cointegration across datasets (Johansen 231; Engle and Granger 251). In equity portfolios, these associations enable pair trading methodologies, capitalizing on short-term deviations from model equilibrium, hence resulting in consistent equity returns (Gatev et al. 802). Nonetheless, across crypto markets, cointegration relationships are transitory, informing regulatory shocks, speculative behaviour and technological disruptions over time (Corbet et al. 28). Alao further reported that as cointegration-based arbitrage techniques have widely been used across developing markets, its observable adaptation to newer digital assets is limited by instability of long-run equilibria and sparsity of data (Alao).

The statistical arbitrage studies inform that fewer studies have explored ML-backed cointegration, in which developed algorithms identify dynamically, evolving equilibrium associations across crypto portfolios. Relevant solutions to this gap call for a blend of adaptive ML and econometric rigor strategies, with the ability of undertaking real-time recalibration, amid any possible market turbulence.

2.3 Pairs Trading in cryptocurrencies trading

Gatev, Goetzmann, and Rouwenhorst (797) was the pioneer for the most ancient pair trading as an arbitrage strategy. This was considered a market-neutral approach, designed to adopt temporary deviations across historically cointegrated or correlated asset pairs. The pair trading approach works under the assumption of mean reversion, in which spread between two assets converge, hence resulting in profitability whenever traders go short on outperforming portfolios and long on underperforming assets. Across traditional foreign exchange (FX) and equity markets, pair trading has been adopted, owing to regulatory stability, market maturity and existence of historical data therein (Lo and MacKinlay). Their limited historical records and heterogeneous liquidity profiles exacerbate difficulties in identifying stable cointegration relationships. Moreover, the absence of centralized oversight or uniform valuation metrics makes fundamental anchoring less reliable, increasing dependence on technical and data-driven methods (Wu et al.).

Nonetheless, inclusion of trading pairs to crypto markets comes with a series of setbacks and opportunities. As opposed to FX or equities, cryptocurrencies exist across decentralized, volatile and 24/7 markets, with periodic regime changes and non-stationarity in return patterns. Their scarce heterogeneous liquidity and historical record profiles outperform the setbacks in the identification of stable cointegration associations (Wu et.al.). Furthermore, the existence of no centralized oversight metrics also makes fundamental approaches less feasible, hence a rise in reliance on data-driven methodologies.

Empirical research has shown that while equities exhibit stable correlations due to shared macroeconomic fundamentals, cryptocurrency correlations fluctuate rapidly due to sentiment-driven movements and evolving network utilities (Vu and Bhattacharyya). Moreover, structural breaks emanating from regulatory announcements, market crashes, or exchange-specific disruptions can invalidate static statistical arbitrage models. Consequently, machine learning (ML) and deep learning (DL) approaches, such as LSTM networks, are increasingly favoured for capturing non-linear dependencies and dynamic cointegration (Hochreiter and Schmidhuber).

Studies have informed that as equities are characterized with stable correlations owing to macroeconomic fundamentals that are shared, cryptocurrencies' correlations change under high frequencies owing to movements driven by sentiments. Moreover, structural breaks emerging from regulatory announcements, market crashes or exchange-related disruptions also invalidate the essence of static statistical arbitrage approaches. Moreover, deep learning (DL) such as LSTM and machine learning (ML) methods such as ensemble models are viable to capture dynamic cointegration and non-linear dependencies (Hochreiter and Schmidhuber 1997).

2.4 ML and DL as new Predictive Trading Strategies

The nexus between DL and ML across financial forecasting has led to more optimistic outcomes across risk-adjusted portfolio optimization, volatility modelling and regime dynamics (Cao et al.). Recurrent Neural Networks (RNNs), especially LSTMs are more viable for sequential data, as they in most cases, outperform linear models, based on how they capture temporal dependencies (Hochreiter and Schmidhuber 1997).

Shah, Vaidya, and Shah (2001) noted that hybrid systems designed using LSTM can expect short-term price movements, with more effectiveness, as opposed to GARCH and ARIMA models. Moreover, ensemble learning approaches which comprise of Bagging, Gradient Boosting and Random Forests, better robustness as they merge diverse learners, limit overfitting and better generalization (Zhou). In the crypto markets, Wu et.al. noted that hybrid LSTM-ensemble architecture realized the best

predictive accuracy, in detection of turning points around the market (Wu et.al.). However, empirical studies are still limited on forecasting tasks, with few of them addressing strategy construction, majorly on long/short pair trading strategies to result in actual actional portfolio allocations from their predictions.

Nonetheless, the past studies evaluated DL and ML models in isolation, with more focus on prediction unlike practical utilization. This project's contribution lies in linking asset selection using cointegration, forecasting (LSTM) and optimal allocation (ensemble methods) for a single trading pipeline for cryptos.

2.5 Competitor Analysis

Various studies have modelled cryptocurrencies using diverse approaches. However, gaps still emerge, especially on the essence of robust approaches to cointegrate, forecast and optimally allocate crypto portfolios as in table 1.

Table 1: Competitor Analysis

Competitor	Strengths	Weaknesses	Opportunities	Threats
CryptoHedge AI (Chen et al.)	Adopts ML for risk management (volatility clustering)	Asset covered is very narrow	Need for multi-asset pair trades	Heightened competition from AI-enabled trading platforms
Shrimpy.io (Kopp and Orlovskyi 274)	Have highly automated portfolio rebalancing tools	Have no DL integrated into their systems	Need to incorporate DL i.e LSTM for their predictions	Extensive regulatory oversight witnessed
Numerai Signals (Konathala)	Extensive uptake of ML (ensemble)	Only focussed mainly on equities	Need to incorporate DL and ML for crypto	Instability of data across financial markets (crypto)
TokenMetrics (Deiseroth et al.)	Does quantitative and fundamental insights	Weaker predictions in the short-term	Need for cointegration approaches	Market volatility is quite a challenge
QuantConnect (Vu and Bhattacharyya)	Works on an open source community; Has more robust rebalancing approaches	Higher learning curve; The set-up is very complex	Introduce ML approaches to crypto markets	Extreme competition from other closed systems

2.6 Gap analysis

Table 2: Gap analysis

Research Gap	Competitors' Limitations	Our Contribution	Citations
Traditional trading strategy setbacks	Linear approaches (buy-and-hold, momentum, equal-weight) does not cater for regime shifts and non-linearities.	LSTM and ensemble data-driven strategies to address capture non-linearity dependencies and market regimes.	(Shah, Vaidya, and Shah 200111))
Static cointegration across cryptos	Johansen and Engle–Granger models often assume static long-run relationships, hence less viable for cryptos.	Uptake of ML-guided cointegration filtering to allow for equilibrium detection across crypto pairs.	(Piallini)
Risk-adjusted portfolio optimization	Existing DL models are built around price prediction with no backtesting or allocation across trading costs in the real-world allocations.	Adopting ensemble risk-adjusted allocations examined using Sortino and Sharpe ratios.	(Zhou)

2.7 Literature review summary

The examined literature reports the absence of integrated frameworks, with the ability to translate predictive insights to systematic pair-trading methodology for cryptocurrencies. Traditional methods failed under non-linearity and high volatility instances (Qureshi). Moreover, the existing limited empirical validation of such frameworks around crypto markets inform the essence of periodic systematic backtesting, cross-validation and comparative evaluation connected to real data, in line with the project's objectives. Newer methodologies such as ensemble and LSTM networks present room to better risk-adjusted returns, predictive accuracy and robustness in response to regime changes. This project was aimed at addressing the noted gaps based on a proposed hybrid methodology, merging cointegration, LSTM forecasting and ensemble-based allocations (Bilgili 134). This multi-tiered approach was aimed at enhancing both empirical and methodological discourse through data-driven and more dynamic solutions viable for statistical arbitrage on cryptocurrencies, beyond fragmented and static models frequently used in contemporary studies.

3.0 Methodology

3.1 Data Sources

The project data was extracted from Yahoo finance, leveraging on nine (9) years of daily historical data for cryptocurrency assets (Bitcoin, Ethereum, Ripple, Litecoin, Cardano, Polkadot, Solana, Avalanche and Binance Coin) from 30th September 2016 to 29th September 2025.

3.2 Data Analysis

The first step was data pre-processing, to log-transform the crypto prices since they are high-frequency volatility signal assets.

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \quad (4)$$

In this case, r_t is the log-returns at price P_t and time t , which is the asset's closing price.

Liquidity screening was adopted to exclude assets with low trading activity. This entailed a 30-day average volume filter, to allow for inclusion of liquidity pairs, which were suitable for real-world trading.

$$\bar{V}_{t,30} = \frac{1}{30} \sum_{i=0}^{20} V_{t-i} \quad (5)$$

In this case, only assets that satisfy $\bar{V}_{t,30} > \theta_V$, whereby θ_V denotes some liquidity threshold were adopted.

The third stage entailed identification of potential trading pairs. This involved a long-run equilibrium relationship which was done using Engle-Granger cointegration test (Engle et.al 250). Given asset pairs (i, j) ;

$$P_{i,t} = \alpha + \beta P_{j,t} + \varepsilon_t \quad (6)$$

Where $\varepsilon_t \sim I(0)$ for cointegration to exist. Pairs which satisfy this condition were viable or long/short term strategy investing.

In capturing non-linear temporal dependencies across cryptocurrency returns, LSTM model was used for forecasting 25-day ahead returns. The process involved the following hidden state and cell state update equations:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (7)$$

$$i_i = (W_i \cdot [h_{t-1}, x_t] + b_i) \quad (8)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (9)$$

$$C_t = f_t \odot C_{t-1} + i_i \odot \tilde{C}_t \quad (10)$$

$$h_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \odot \tanh(C_t) \quad (11)$$

In which f_t , i_i and o_t being the forget, input and output gates for LSTM model.

The last step was prediction of the returns from LSTM using ensemble learning models (Random Forest (RF), Gradient Boost (GB) and Linear Regression (RG)), for constructing a risk-adjusted portfolio allocation. The ensemble forecast was defined s;

$$\hat{r}_t = w_{LSTM} \widehat{r_{t,LSTM}} + w_{RF} \widehat{r_{t,RF}} + w_{GB} \widehat{r_{t,GB}} \quad (12)$$

Subject to

$$\sum_i w_i = 1, w_i > 0$$

The weights w_i undergo optimization on out-of-sample performance, for maximizing expected risk-adjusted returns for the cryptos.

3.3 Model Evaluation

Model performance was done by benchmarking against the baseline strategies (Buy-and-Hold, Momentum and Equal-Weight) strategies. The evaluation metrics comprised of:

Cumulative returns:

$$R_{cum} = \prod_{t=1}^T (1 + r_t) - 1 \quad (13)$$

Sharpe ratio:

$$SR = \frac{\bar{r}_p - r_f}{\sigma_p} \quad (14)$$

Sortino ratio:

$$SoR = \frac{\bar{r}_p - r_f}{\sigma_d} \quad (15)$$

Momentum drawdown:

$$MDD = \max_{t \in [0, T]} \frac{P_{peak} - P_t}{P_{peak}} \quad (16)$$

Hit ratio:

$$HR = \left(\frac{\text{Number of correct predictions}}{\text{Total predictions}} \right) \quad (17)$$

Net return after transaction costs:

$$R_{net} = R_{gross} - c \times N_{trades} \quad (18)$$

Where r_f is the risk-free rate, c is cost per trade, σ_d is downside deviation, σ_p is portfolio deviation, N_{trades} is total number of transactions done.

4.0 Results

4.1 Exploratory Data Analysis

The summary statistics informed that Bitcoin (BTC-USD) had the highest average price and volatility ($M=\$49,327$, $SD=\$28,735$) as Cardano had the least average prices ($M=\$0.71$, $SD=\$0.53$). Ethereum (ETH-USD) ($M=\$2,340$, $SD=\$1,014.16$) had a moderate dispersion. Avalanche (AVAX-USD) and Solana (SOL-USD) (Altcoins) reported reasonable variability, informing instability across these markets. The notable high volatility across the cryptocurrency markets were witnessed from high price ranges as reported in Table 3 and Figure 1.

Table 3: Descriptive statistics for crypto prices

Ticker	BTC-USD	ETH-USD	XRP-USD	LTC-USD	ADA-USD	DOT-USD	SOL-USD	AVAX-USD	BNB-USD
count	1867.00	1867.00	1867.00	1867.00	1867.00	1867.00	1867.00	1867.00	1867.00
Mean	49326.75	2339.94	0.91	102.94	0.71	11.29	88.59	29.88	400.01
Std	28735.44	1014.16	0.78	50.17	0.53	10.65	72.96	23.52	207.24
Min	10131.52	321.12	0.21	43.06	0.08	2.88	1.21	2.91	19.48
25%	26826.09	1640.44	0.46	68.81	0.35	4.67	22.18	14.86	261.76
50%	42270.53	2241.98	0.58	87.83	0.51	6.40	62.43	22.92	344.18
75%	64078.28	3117.95	0.99	119.65	0.92	13.79	150.86	35.97	580.78
Max	123344.06	4831.35	3.56	386.45	2.97	53.88	261.87	134.53	1048.67

Sample period: 30th September 2016 to 29th September 2025

Figure 1 reported that Bitcoin reported the highest closing prices for the past 9-years, when compared to other cryptos.



Figure 1: Time series of daily closing prices

Sample period: 30th September 2020 to 29th September 2025

Note: Some cryptos had no observations before 2020 hence the whole sets were excluded.

The trends for returns in Figure 2 informed that the average daily returns were mostly positive, though smaller with ranges from 0.13% for LTC to 0.42% for SOL, an indication of modest growth trends. Nonetheless, the reported higher volatilities of 3.09% on BTC and >0.6% for SOL and AVAX showed significant volatility. Additionally, extreme negatives such as -37.93% for DOT and -42.33% for XRP also informed periods with sharper sell-offs, as maximum returns of 75% for AVAX showed speculative margins. The trend reported that the returns showed high risk high reward tendencies for the different cryptos.

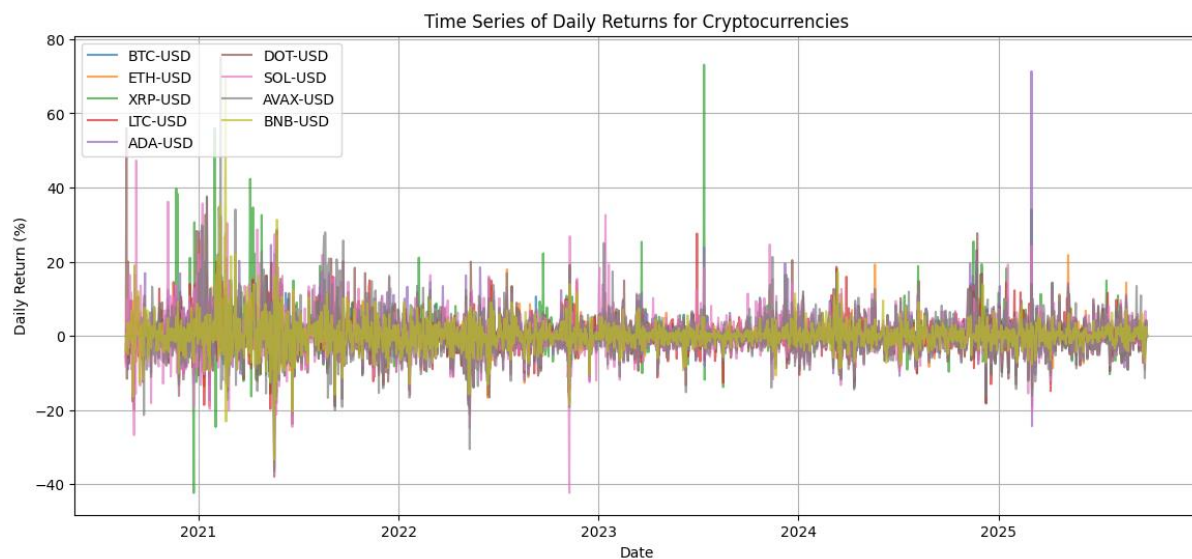


Figure 2: Time series of daily returns

Sample period: 30th September 2020 to 29th September 2025

Note: Some cryptos had no observations before 2020 hence the whole sets were excluded.

Figure 3 revealed that volatility was at its peak in 2020, showing heightened market speculation, which decreased during 2022-2023 periods, with lower surges. Cryptos like AVAX, ADA and SOL had higher and erratic volatility as opposed to ETH and BTC which were more sensitive to market shocks. BTC was relatively stable across the 9-years, hence least volatile from their volatility (%) indices, when

compared for the rest of the crypto assets, across the 9-year periods.

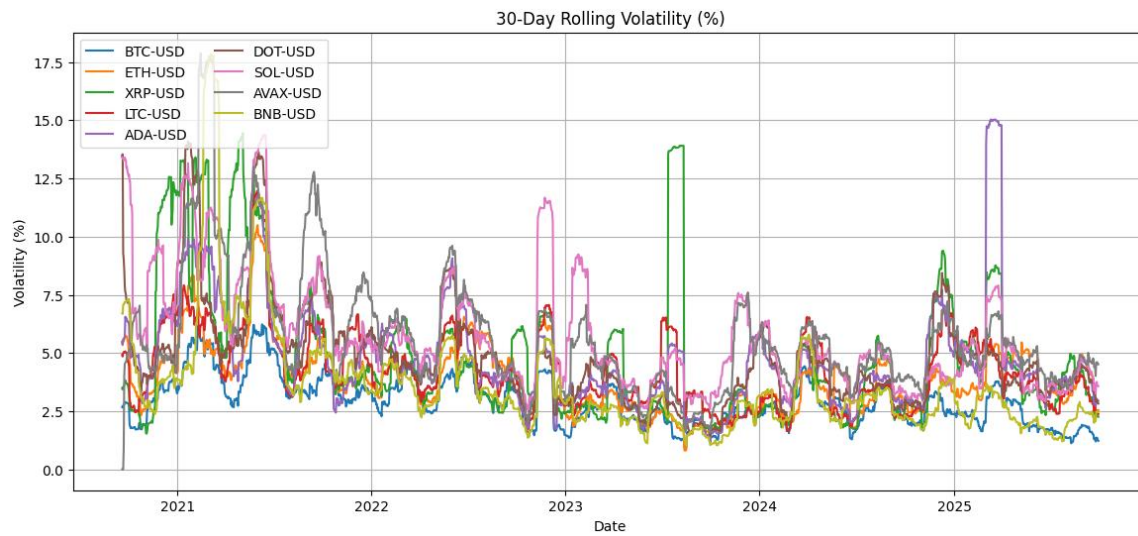


Figure 3: 30-Day rolling volatility (%)

Sample period: 30th September 2020 to 29th September 2025

Note: Some cryptos had no observations before 2020 hence the whole sets were excluded.

4.2 Cointegration and pair trading

The cointegrated and pair trading in Figure 4 only found one (1) cointegrated pairs (LTC-USD and DOT-USD) ($p < .05$) as statistically significant. This explained a stable long-term equilibrium relationship between these asset prices. The outcome showed that the deviations between these asset prices were likely to revert with time or temporary, hence viable for pair trading in terms of mean reversion.

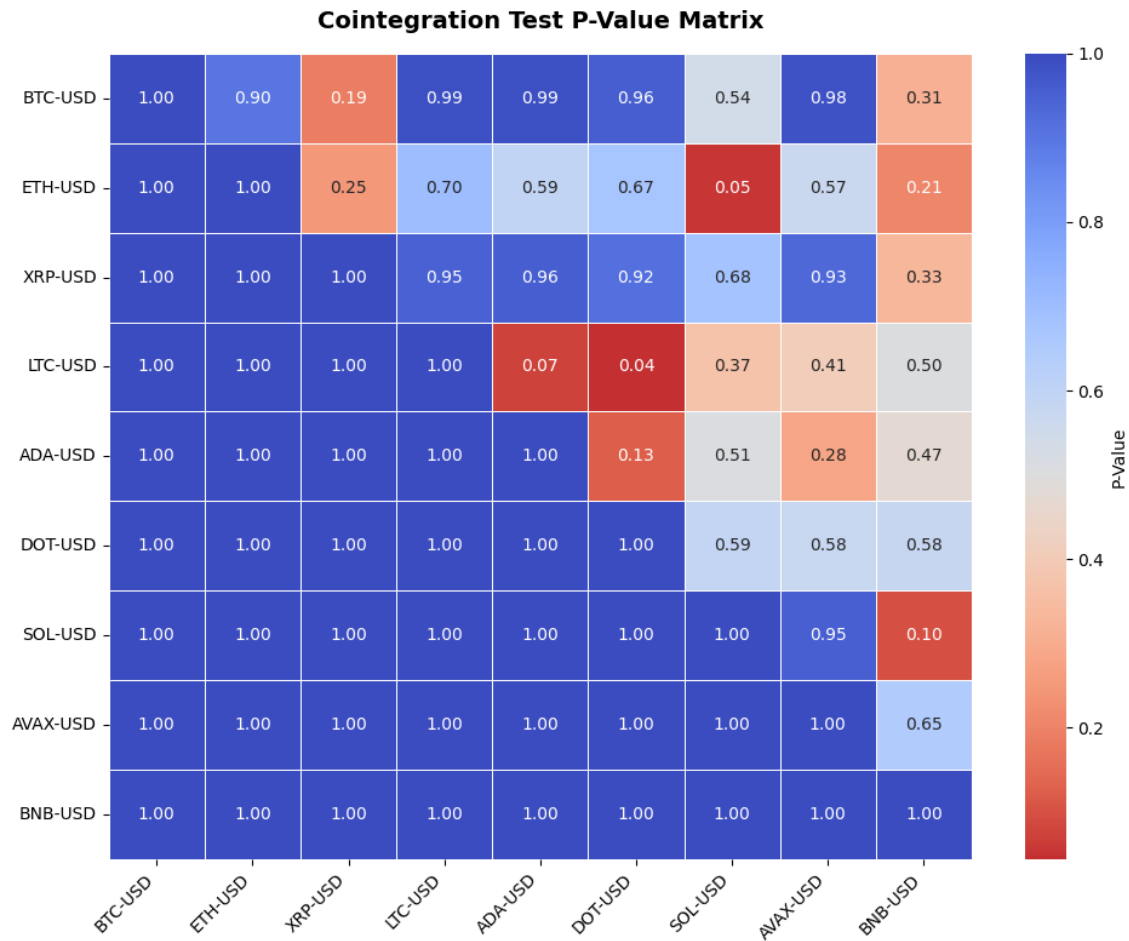


Figure 4: Cointegration and pair trading

The calculations used in figure 4 were based on the crypto returns, to attain the desired cointegration and pair-trading outcomes for the different crypto portfolios used.

4.3 Random Forest (baseline model) Allocation

Training-testing was done at 80%-20%, hence the modelling period done from 2023-2025, owing to the missing values reported for ETH, XRP, LTC, ADA, DOT, SOL, AVAX and BNB.

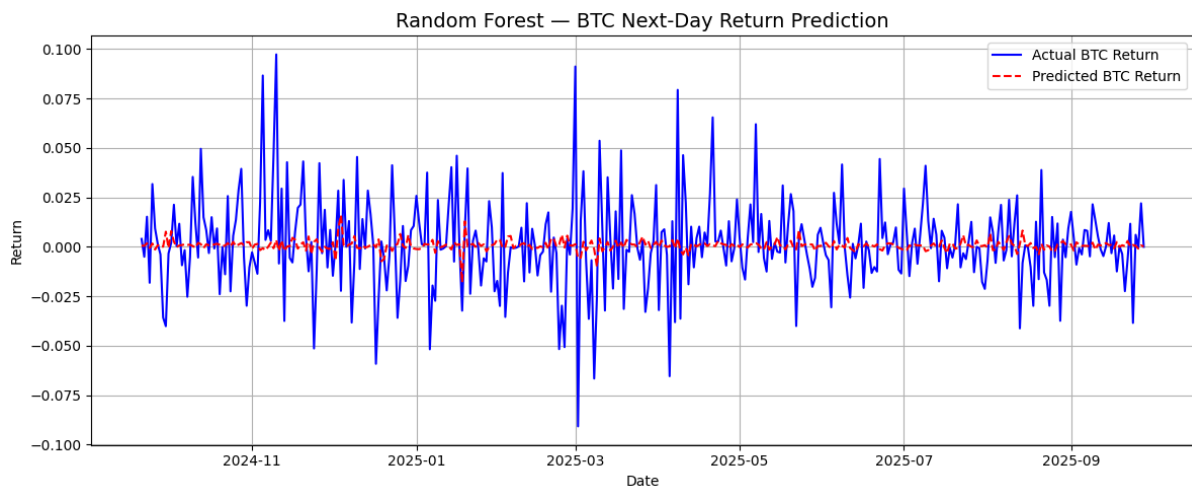


Figure 5: Random Forest (Next day predictions)

Sample period: 30th September 2024 to 29th September 2025

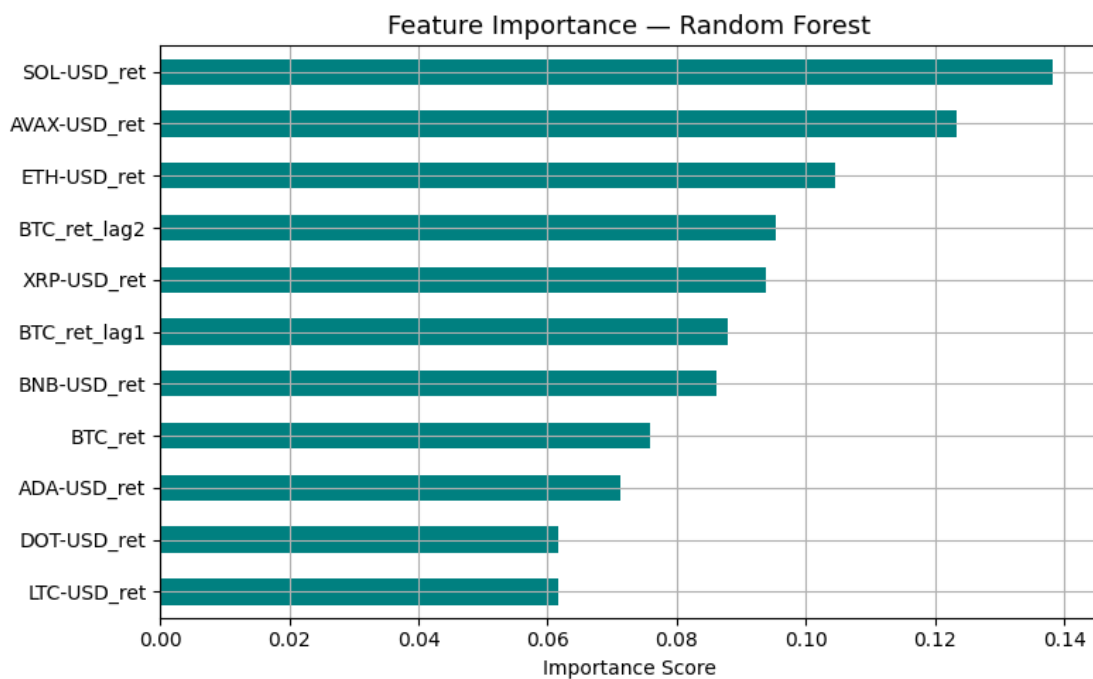


Figure 6: Random Forest Feature Importance

SOL-USD, AVAX-USD, ETH-USD, BTC (lag 2), XRP-USD, BTC-USD (lag 1), BNB-USD, BTC-USD, ADA-USD, DOT-USD, LTC-USD was the order of feature importance from the model (Figure 6). The Random Forest model showed that SOL-USD, AVAX-USD, and ETH-USD were the most influential predictors of BTC returns. This indicated strong interdependence among major cryptos. Nonetheless, the low R-squared value of 0.0003 and MSE of 0.00052 informed that the model explained very minimal variation, hence weak predictive accuracy.

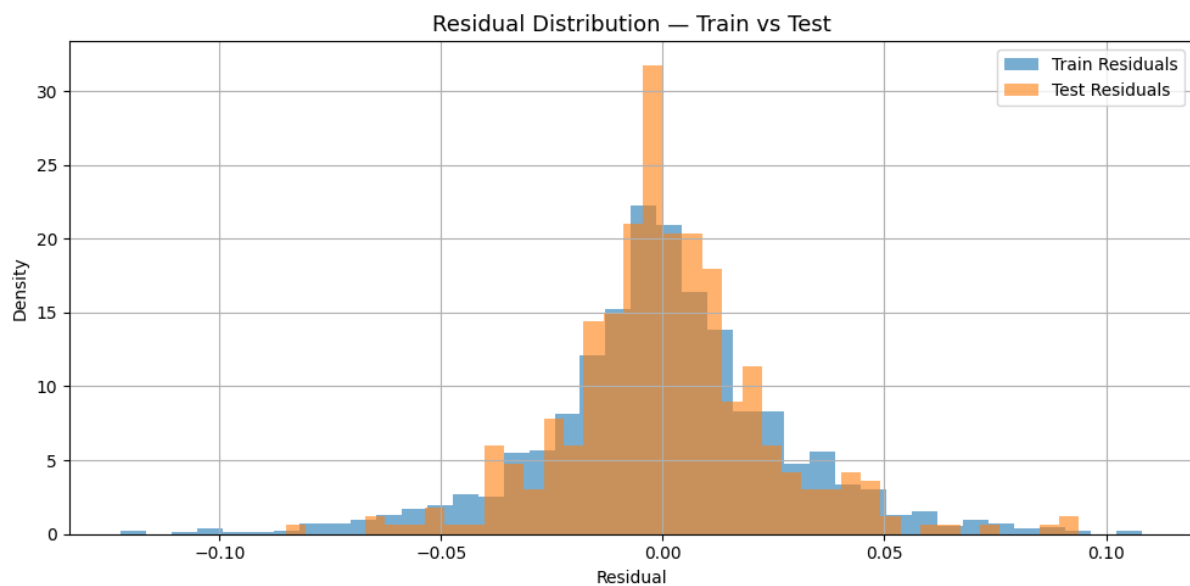


Figure 7: Random Forest overfitting checks

The overfitting checks showed that OOB R-squared score was -0.0206 in Figure 7. The residual variance for training was 0.000771 while test residuals was 0.000520, hence showing some levels of overfitting, despite residuals being consistent across train/test samples, hence no severity on the overfit.

4.4 Ensemble Learning Allocation

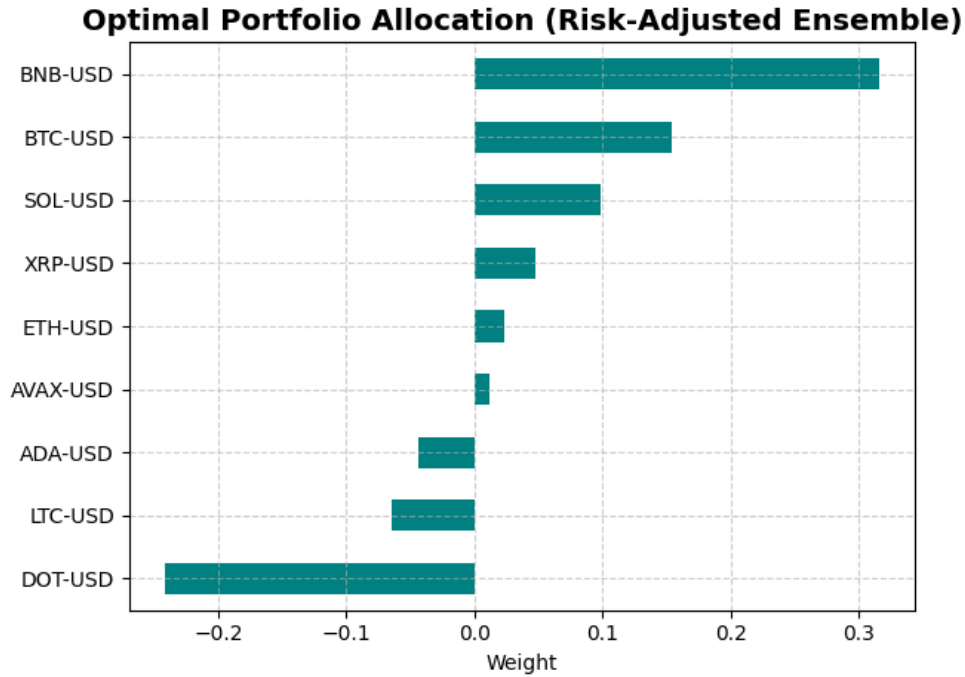


Figure 8: Optimal Portfolio Allocation (Ensemble)

Sample period: 30th September 2024 to 29th September 2025

Table 4: Ensemble mean predicted return and optimal weights

Crypto	Predicted Return (μ)	Optimal Weight
BNB-USD	0.001430	0.3158
BTC-USD	0.001039	0.1535
SOL-USD	0.002232	0.0985
XRP-USD	0.000678	0.0472
ETH-USD	0.001174	0.0236
AVAX-USD	0.000600	0.0112
ADA-USD	0.000462	-0.0435
LTC-USD	-0.000445	-0.0648
DOT-USD	-0.000971	-0.2418
Expected return 0.1141%		
Volatility	0.9704%	
Sharpe ratio	0.1176	

Sample period: 30th September 2024 to 29th September 2025

The ensemble model in Figure 8 and Table 4 optimized the portfolio through allocation of higher weights to low volatility, high return assets like SOL-USD and BNB-USD, as they assigned negative weights to underperforming cryptos. This was done for hedging purposes, such as on LTC-USD and DOT-USD. Despite the witnessed moderate volatility (0.9704%), the Sharpe ratio of 0.1176 informed limited risk-adjusted performance, hence the essence for diversification to better predictive outcomes and enhance efficiency on the crypto portfolios.

4.5 LSTM forecasting performance

Table 5: LSTM model summary

Model: "sequential_10"

Layer (type)	Output Shape	Param #
lstm_18 (LSTM)	(None, 30, 64)	18,944
dropout_18 (Dropout)	(None, 30, 64)	0
lstm_19 (LSTM)	(None, 64)	33,024
dropout_19 (Dropout)	(None, 64)	0
dense_18 (Dense)	(None, 24)	1,560
dense_19 (Dense)	(None, 1)	25

Total params: 160,661 (627.59 KB)
Trainable params: 53,553 (209.19 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 107,108 (418.39 KB)

The LSTM model was comprised of two LSTM layers having 64 units each with a dropout regularization in table 5. This was later followed by two dense layers adopted for forecasting performance. The model comprised of 160,661 parameters, of which 53,553 were trainable, majorly from the LSTM weights as 107,108 were from the optimizer state with a learning rate of 0.001.

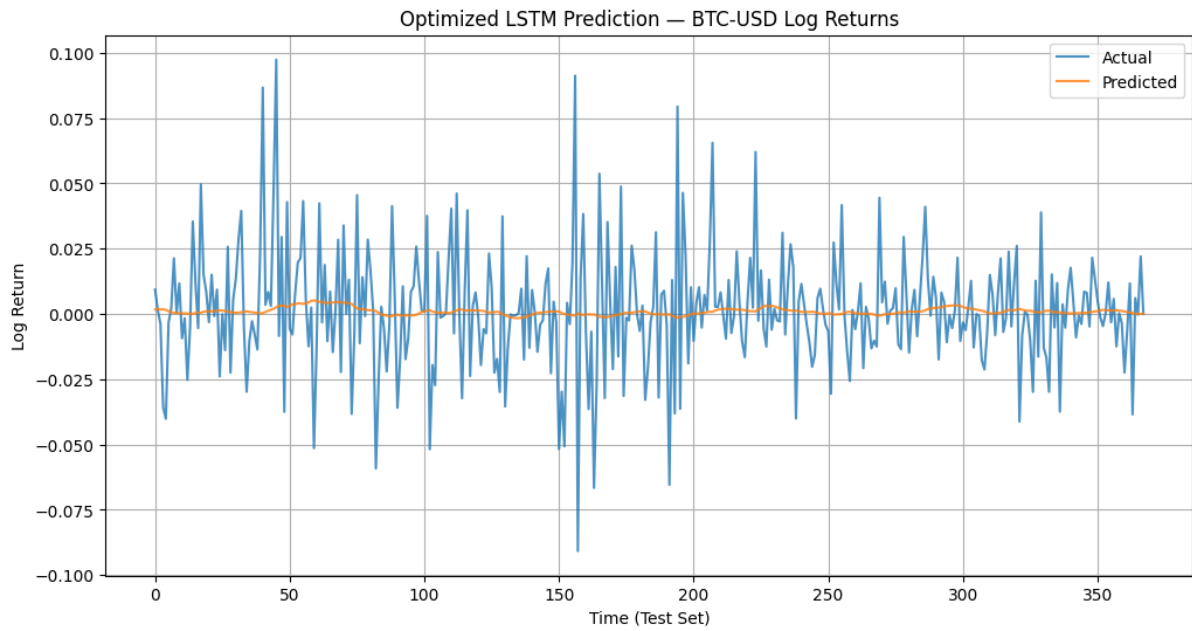


Figure 9: Optimized LSTM predictions (BTC-USD)

Sample period: 30th September 2024 to 29th September 2025

The findings from Figure 9 reported highly volatile log returns, as the model predictions capturing a general trend, which fails to track sharp spiles. The small model's MSE showed small average prediction with a worse predictive power ($R\text{-Squared}=-0.0347$). The model captured wider patterns, but reported struggles with extreme volatility, which is a major setback with crypto forecasting.

4.6 Backtesting and strategy evaluation

The model performance in Figure 12 and Table 3 revealed that LSTM was consistent in outperforming both Random Forest and Ensemble models, across Buy-hold, momentum and equally-weighted strategies. On equal-weighted strategy, LSTM had the highest net return of 0.784 and a Sharpe ratio of 0.91, hence an indication of risk-adjusted returns. Despite Random Forest and Ensemble performing moderately under buy-hold, their returns were always lower. Moreover, LSTM also reported stronger downside protection, owing to the lower maximum drawdown reported (-0.43), making LSTM superior on risk management and profitability across all provided strategies.

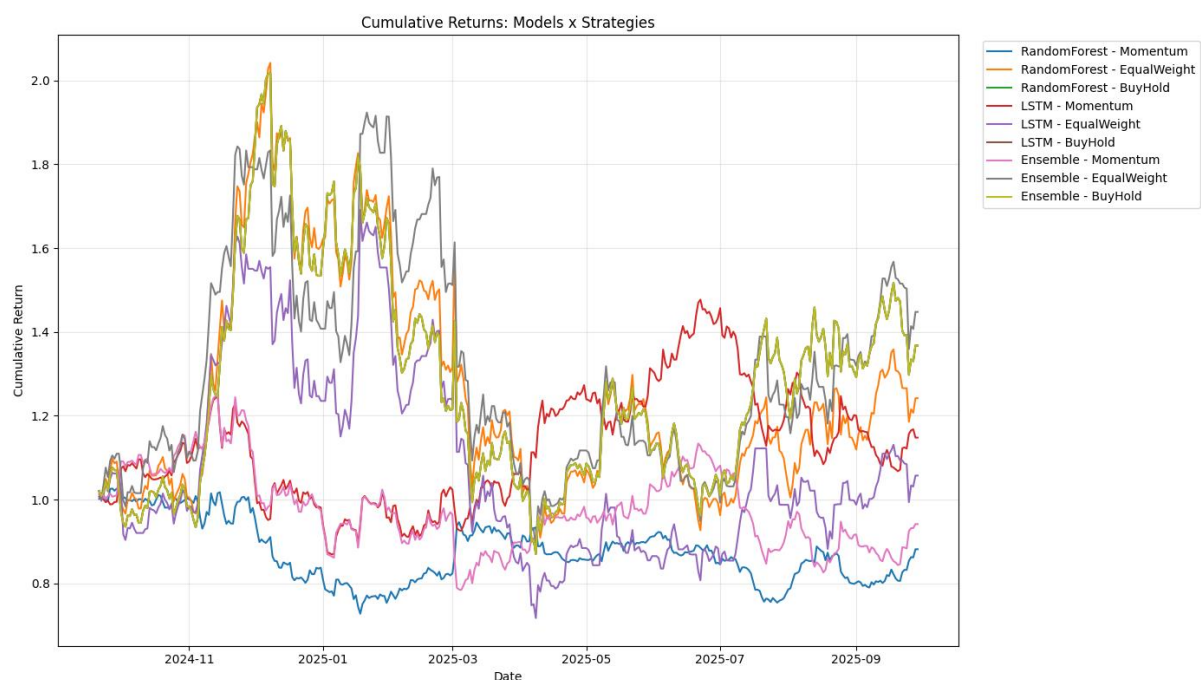


Figure 10: Backtesting (cumulative returns: Models and their strategies)

Sample period: 30th September 2024 to 29th September 2025

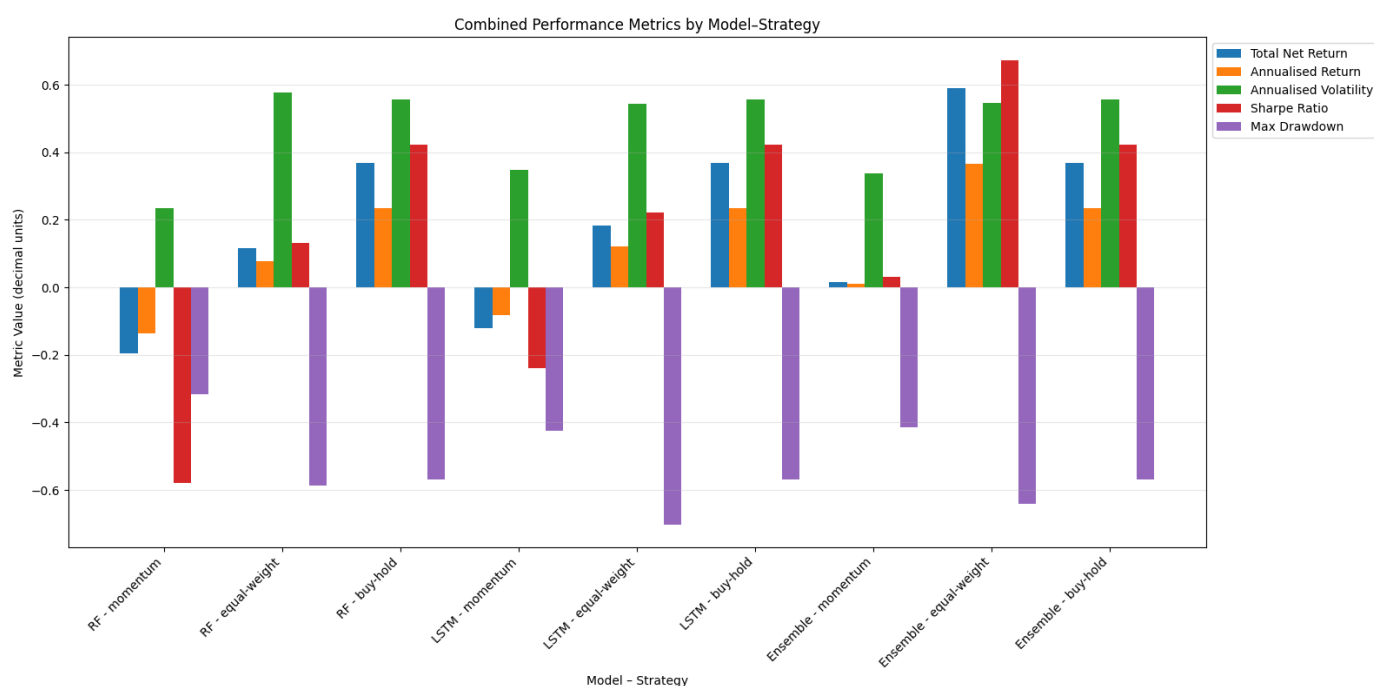


Figure 11: Model-Strategy performance

The figure plots performance metrics for total net return, annualized return, annualized volatility, Sharpe ratio and max Drawdown for the RF, LSTM and ensemble models. It also includes equal-weight, buy-hold and momentum trading strategies on the cryptos.

The analysis revealed that Ensemble+Equal-weight had the strongest returns and best risk-adjusted metrics. Buy and hold approach was solid and consistent, hence showing the best passive benchmark across the market. The weakest strategy was RF momentum, as it was associated with a low Sharpe ratio and negative returns. For the three models, Ensemble was the strongest, as LSTM proved moderate as RF was the weakest.

5.0 Discussions

5.1 Discussion of the findings

The cointegration test revealed one statistically significant pair LTC-USD and DOT-USD. This confirmed the evidence of a stable long-term equilibrium, desirable for mean-reversion trading. The outcome informed that the deviation on price relationships were short-lived, hence validating room for pair trading approaches.

The Random Forest model reported that SOL-USD, AVAX-USD, and ETH-USD were the most influential predictors of BTC returns over the 9-year periods. However, its low R^2 of 0.0003 and MSE of 0.00052 inferred a weak predictive power. Moreover, the overfitting checks confirmed meagre training-test discrepancies, limiting its reliability for trading decisions.

The Random Forest model showed that the most influencers of BTC returns were SOL-USD, AVAX-USD, and ETH-USD. Nonetheless, the reported low R-Squared of 0.03% and MSE of 0.00052 showed a lower predictive power. Moreover, there was evidence of overfitting on the training set, hence limiting trading decisions, based on the model.

The Ensemble model enhanced portfolio allocation through assigning oof higher weights to low-volatility and high-return assets such as SOL-USD and BNB-USD and negative weights to low performers such as DOT-USD and LTC-USD to hedge against losses. Despite the witnessed moderate volatility of 0.97%, the low Sharpe ratio of 0.1176 indicated limited risk-adjusted performance, hence the essence of diversification to better the portfolio outcomes' predictability.

The LSTM model, despite having issues with sharp spikes on the cryptos and lower predictive power, effectively captured wider trends. Backtesting reported that LSTM outperformed ensemble and Random Forest models, to realize the highest net returns of 0.784, a Sharpe ratio of 0.91 and the least maximum drawdown of -0.43, hence reported a more robust and superior risk-adjusted performance in crypto trading.

5.2 Possible hurdles

The study faced a challenge of data availability, since most cryptos could not yield data for the 9 years from 2016, especially ETH, XRP, LTC, ADA, DOT, SOL, AVAX and BNB, which made modelling commence in 2020, not the anticipated 2016. This limited the study's scope, a challenge which can be offset in the coming years, say 2030 or 2040 where there will be robust longitudinal and time series data, relevant for crypto forecasting.

A major setback witnessed from the project was the existence of frequent structural breaks and extreme volatility across crypto markets. Price behaviour across digital assets was highly sensitive to technological upgrades, market sentiment, macroeconomic shocks and regulatory announcements among others. These stochastic shifts could disrupt stability assumptions around deep learning models and cointegration testing. In this case, better performing models at stable periods may prove unreliable during unstable regimes, hence affecting the viability of some models, across diverse market cycles.

5.2 Real Life Application

The superior performance of LSTM-guided forecasting signals alongside their higher Sharpe ratios makes them viable for risk-adjusted portfolio optimization. Robo advisers, crypto exchanges and other fintech investment platforms can include LSTM algorithms to their diverse trading engines for optimizing asset allocation around volatile tokens such as DOT, SOL, LTC and AVAX. These models would provide room for automated systems detect changing price trends, as opposed to earlier machine learning models, to allow for timely and more dynamic rebalancing, to reduce downside losses whenever markets experience drawdowns.

Moreover, the study informed the merits associated with ensemble methods in identifying low-volatility, high return assets can be included in ETFs and crypto index funds. For instance, digital asset managers could include Ensemble signals for constructing robust equal-weighted crypto indices which outperform basic market standards. Inclusion of these models to real-time crypto trading dashboards, institutional players and investors can aid in risk evaluation, allow for prediction of momentum and further aid in hedging underperforming coins efficiently. The results strongly support real-world inclusion of real-world deployment of machine and deep learning as a component of automated trading system, quantitative risk management frameworks and portfolio rebalancing to expand digital asset portfolios and markets.

6.0 Conclusions

This study informed the relevance of both ML and DL approaches in trading cryptocurrency pairs and portfolio management. Cointegration analysis noted DOT-USD and LTC-USD as the only significant pair, viable for mean-reversion strategies. Moreover, Random Forest model, as it highlighted SOL-USD and AVAX-USD as the most important predictors, reported minor overfitting and weak predictive power, hence limiting its effectiveness. Furthermore, Ensemble learning enhanced portfolio allocation by concentrating on low-volatility, high-return assets, hedging underperformers, despite the risk-adjusted performance still remained moderate. LSTM models, despite the setback on capturing sharp price spikes on cryptos, tracked broader trends across the portfolio, hence outperforming Ensemble and Random Forest models, across buy-and-hold, momentum and equal-weighted strategies.

Moreover, backtesting confirmed LSTM strategies as associated with higher Sharpe ratios, superior net returns and downside protection, hence the model more viable for crypto markets, despite their volatility. This informs that deep learning techniques offer more efficient portfolio optimization approaches, and better predictive accuracy, unlike ML model.

Future studies can adopt a mix of deep learning methods such as Recurrency Neural Networks (RNN) and Convolution Neural Networks (CNN) alongside LSTM, in cryptocurrency trading, to determine the model with the best forecasting performance, to better trading. This need to be done, in line with momentum, equal-weighted and buy-hold strategies to inform trading using real-time algorithms.

7.0 Bibliography

1. Acharya, Paru. "An R-Based Analysis of a Moving Average Crossover Strategy vs. Buy-and-Hold: Evidence from Several Major Tech Stocks." *Buy-and-Hold: Evidence from Several Major Tech Stocks (March 20, 2025)* (2025).
2. Alao, Oluwatobiloba. "Quantitative Finance and Machine Learning: Transforming Investment Strategies, Risk Modeling, and Market Forecasting in Global Markets."
3. Bilgili, Faik. "Stationarity and cointegration tests: Comparison of Engle-Granger and Johansen methodologies." *Erciyes Üniversitesi İktisadi ve İdari Bilimler Fakültesi Dergisi* 13 (1998): 131-141.
4. Cao, Xinwei, et al. "Neural networks for portfolio analysis in high-frequency trading." *IEEE Transactions on Neural Networks and Learning Systems* (2023).
5. Chen, Jun-Hao, et al. "Financial vision based differential privacy applications." *arXiv preprint arXiv:2112.14075* (2021).
6. Corbet, Shaen, et al. "Exploring the dynamic relationships between cryptocurrencies and other financial assets." *Economics letters* 165 (2018): 28-34.
7. Deiseroth, Björn, et al. "Divergent Token Metrics: Measuring degradation to prune away LLM components--and optimize quantization." *arXiv preprint arXiv:2311.01544* (2023).
8. Engle, Robert F., and Clive WJ Granger. "Co-integration and error correction: representation, estimation, and testing." *Econometrica: journal of the Econometric Society* (1987): 251-276.
9. Eugene, Fama. "Efficient capital markets: A review of theory and empirical work." *Journal of finance* 25 (1970): 383-417.
10. Ferrouhi, El Mehdi, and Ibrahim Bouabdallaoui. "A comparative study of ensemble learning algorithms for high-frequency trading." *Scientific African* 24 (2024): e02161.
11. Fior, Jacopo. "Machine Learning-Driven Decision Making based on Financial Time Series." (2023).
12. Gatev, Evan, William N. Goetzmann, and K. Geert Rouwenhorst. "Pairs trading: Performance of a relative-value arbitrage rule." *The review of financial studies* 19.3 (2006): 797-827.
13. Hochreiter, Sepp, and Jürgen Schmidhuber. "Long short-term memory." *Neural computation* 9.8 (1997): 1735-1780.
14. Hu, Y., et al. "Deep learning for financial time series forecasting: A survey." *2018 IEEE International Conference on Progress in Informatics and Computing (PIC)*. IEEE, 2018.
15. Johansen, Søren. "Statistical analysis of cointegration vectors." *Journal of economic dynamics and control* 12.2-3 (1988): 231-254.
16. Konathala, Lohith. "Open Innovation in Hedge Funds: Collective Intelligence Investing Using Crowdsourcing." *Available at SSRN 4834730* (2023).
17. Kopp, Andrii, and Dmytro Orlovskyi. "Towards the Tokenization of Business Process Models using the Blockchain Technology and Smart Contracts." *CMIS* 3137 (2022): 274-287.
18. Lo, Andrew W., and A. Craig MacKinlay. "A non-random walk down Wall Street." *A Non-Random Walk Down Wall Street*. Princeton University Press, 2011.
19. Mikolajek-Gocejna, Magdalena. "Estimation, instability, and non-stationarity of beta coefficients for twenty-four emerging markets in 2005-2021." (2021).
20. Piallini, Edoardo. "Pairs trading strategies: a cointegration-based approach."
21. Qureshi, Ghazia Khoulia. *Extreme Value Behavior in Cryptocurrency Market*. Diss. Capital University, 2023.
22. Stanciu, M. A. R. I. A. "Artificial Intelligence-Driven Systems for Financial Forecasting." (2022).

23. Vu, Duc Long, and Ritabrata Bhattacharyya. "Design and Development of Mean Reversion Strategies on QuantConnect Platform." *Available at SSRN 4878676* (2024).
24. Wu, Jingyang, et al. "Review of deep learning models for crypto price prediction: implementation and evaluation." *arXiv preprint arXiv:2405.11431* (2024).
25. Zhou, Zhi-Hua. *Ensemble methods: foundations and algorithms*. CRC press, 2025.

8.0 Appendices

Appendix I: Code Repository

The results were computed and implemented using Python, hosted in Github. The components of the notebook are:

1. Environment Set-up
2. Cryptocurrency data extraction
3. Exploratory Data Analysis
4. Cointegration test (Pair Trading)
5. Feature engineering
6. Random Forest Modelling
7. LSTM deep learning modelling
8. Ensemble model (risk-adjusted measures)
9. Backtesting (Long/Short Pairs)
10. Evaluation Metrics

Appendix II: Access instructions

The codes used will be available on written request to authors.

Appendix III: License

This notebook is released under the MIT License, permitting academic and commercial reuse with attribution.